Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Open-Source AI Tools · Agentic Systems · Python · 0-to-1 Builder · LLM Evaluation · Production AI

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

SUMMARY

Software engineer with 9+ years of production engineering depth and a pattern of building AI tools from scratch that solve real operational problems. Built **AgentProbe** - an open-source LLM evaluation toolkit - because I needed a way to know whether my production AI systems were degrading quietly: CI-integrated regression harness, LLM-as-judge scoring, 5-metric eval engine, multi-provider routing. Same category as Imbue's Vet. Built **Leasel** from a blank page: chose the architecture, designed the schema, shipped three production surfaces sharing one knowledge repository (8-agent LangGraph system with HITL escalation, OCR-fallback RAG, and a CI regression gate). Both projects are on GitHub. At Entrata (9+ years, 100K+ users), owned production systems end-to-end across the full SDLC - shipped 20+ modules, led microservices redesign, owned incident resolution and postmortems. Strong Python, FastAPI, LangGraph, RAG pipelines, REST APIs, AWS, Docker, CI/CD. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Python · FastAPI · Django | Open-Source Development | 0-to-1 Product Building | Agentic AI Systems | LLM Evaluation · CI Regression | LangGraph · LangChain | RAG Pipelines · Hybrid Search | REST APIs · OpenAPI | PostgreSQL · SQLite · Redis | AWS (S3 · EC2 · RDS · Lambda) | Docker · Kubernetes | CI/CD · GitHub Actions | Distributed Systems | Multi-Tenant Architecture | TypeScript · Node.js · JavaScript | Java · Spring Boot | HITL · Guardrails · Audit Logging | Production Incident Ownership

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, data pipelines, and AI-powered application features end-to-end; connecting AI components to backend APIs and surfacing insights into reporting layers.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built an agentic mission-planning system (LangGraph StateGraph) - prototype to deployed integration: RAG over regulatory documents, validation guardrails, conditional routing to export or rejection; deployed MCP server connecting agent outputs to downstream enterprise tooling.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming pipeline from regulatory corpus to real-time operator interface.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Shipped 20+ full-stack production modules end-to-end** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints - database schema through API design through deployment pipelines; owned each one in production after launch.
- **Led production incident resolution and postmortems** across services - built internal debugging tools that cut debugging time by 40%; drove root-cause analysis and resiliency improvements; maintained bug count below 10 for 6+ consecutive months across a platform serving 100K+ users.
- **Built and operated distributed async processing** (RabbitMQ) with retry logic and idempotency; owned annual HUD compliance pipeline (few thousand federal rows → ~180,000 domain records) for 7 years, zero compliance violations; PHI-encrypted AWS S3 migration (100% coverage, zero security incidents).
- Optimized 200+ SQL queries (30% gain) across complex multi-tenant PostgreSQL schemas; led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction; Python/Bash automation eliminated 80% of manual operational workload.
- Mentored 5+ engineers; conducted 300+ peer code reviews; drove engineering standards across a team shipping in a regulated environment.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration from legacy storage; delivered large-scale data import workflows (1.6M records).

PROJECTS

AgentProbe - Open-Source LLM Evaluation Toolkit

2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Built to solve a real problem: knowing whether a production AI system is degrading before users notice. **CI-integrated regression harness** with LLM-as-judge scoring and golden datasets; 15-endpoint FastAPI backend; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Used daily as a development tool - the same category as Imbue's Vet.

Leasel - Agentic AI Platform Built 0-to-1

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Took the problem from blank page to production: chose the architecture (LangGraph StateGraph), designed the schema, built three surfaces sharing one SQLite knowledge repository - **document intelligence** (OCR, hybrid search, citation grounding), **agentic investigation** (8-agent pipeline with HITL escalation when confidence is insufficient), and **self-service** with a hard safety boundary on regulated decisions. CI regression gate with AgentProbe LLM-as-judge scoring; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017